

Revision Notes for Class 8 Maths Chapter 1: Fractions as Percentages

In day-to-day life, we often come across the term “per cent” while reading news, hearing about discounts, or discussing exam scores. The symbol % is read as “per cent”, a short form for “per hundred”. For example, 25% means 25 out of every 100, while 50% means 50 out of every 100. This makes it easy to compare quantities of different sizes by fixing a common base of 100.

Fractions and percentages are closely related. Any fraction can be expressed as a percentage by converting it so that its denominator is 100. For example, $\frac{3}{4}$ is equivalent to $\frac{75}{100}$, so $\frac{3}{4}$ is 75%. Likewise, $\frac{2}{5}$ can be rewritten as $\frac{40}{100}$, or 40%. To convert a fraction into a percentage, multiply it by 100. Similarly, any percentage can be turned into a fraction by writing it over 100, such as $24\% = \frac{24}{100}$.

Expressing Fractions as Percentages

Let's look at some examples. Suppose red paint makes up $\frac{3}{4}$ of a mixture—in percentage, this is 75%, since $\frac{3}{4} = \frac{75}{100}$. This can be found by writing the fraction with a denominator of 100, or by calculating $(\frac{3}{4}) \times 100 = 75$. Similarly, if someone saves $\frac{2}{5}$ of their prize money, that equals 40%, by computing $(\frac{2}{5}) \times 100 = 40$. To express a percentage as a fraction, place the number over 100 and simplify if possible; for example, 24% becomes $\frac{24}{100} = \frac{6}{25}$.

Comparison and Uses of Percentages

Percentages make comparisons easier because different quantities can be measured on a common base. For instance, suppose two biscuit varieties contain sugar in the ratios $\frac{9}{34}$ and $\frac{13}{45}$. Converting these to percentages (about 26.47% and 28.88% respectively) quickly reveals which is more sugary. This is much simpler than comparing the original fractions.

Percentages are also used to describe changes, such as profit, loss, or taxes. If the price of something increases or decreases, the difference can be shown as a percent of the original. Interest rates on money are also expressed as percentages, and can be either simple (where only the original 'principal' is used to calculate interest each time) or compound (where interest is added to the principal for subsequent calculations).

Real-life Examples and Applications

Understanding percentages helps in many real-life situations:

- The human body contains about 60% water by weight.
- Ice cream is about 30–50% air by volume.
- Over 80% of teenagers do not get the recommended daily physical activity.
- Sales, discounts, and examination results are always talked about in percentages.
- Mass of the Sun forms about 99.86% of the Solar System's total mass.

Being familiar with percentages helps us better understand information in news, advertisements, and even health reports.

Finding Percentages of Quantities

Often we need to find what a certain percentage of a quantity is, or which percentage a part is of the whole. For example, if Nandini has 25 marbles, of which 15 are white, then $(15/25) \times 100 = 60\%$, so 60% of her marbles are white. Similarly, if 15 out of 80 students come to school by walking, $(15/80) \times 100 = 18.75\%$ walk to school.

In comparisons, it's important to remember not to compare percentages alone if the "whole" is not the same for both cases. For instance, if Madhu's biscuits have 25% sugar and Madhav's have 35%, but they ate different amounts, we must calculate the actual sugar each consumed using the percentages and the total quantity they ate: sugar = $(\text{percentage}/100) \times \text{total grams}$.

Conversion Between Fractions and Percentages

To sum up, the relationship between fractions and percentages is simple:

- To convert a fraction to a percentage, multiply the fraction by 100. Example: $\frac{1}{3}$ as a percentage is $(\frac{1}{3}) \times 100 \approx 33.33\%$.
- To convert a percentage to a fraction, place the number over 100. Example: $30\% = \frac{30}{100} = \frac{3}{10}$.

If a quantity is given in ratio form, we can also express each part as a percentage. For example, if a class contains boys and girls in the ratio 2:3, the percentage of boys is $(\frac{2}{5}) \times 100 = 40\%$; girls is $(\frac{3}{5}) \times 100 = 60\%$.

Solving Problems with Percentages

Many mathematical problems can be solved using different approaches. For instance, to find the answer for what percentage $\frac{9}{20}$ is, we calculate $(\frac{9}{20}) \times 100 = 45\%$. For increase/decrease, the formula for percent change is: $\text{change} = (\text{difference/original amount}) \times 100$.

Percentages can also help us quickly solve “greater than”, “less than”, or “equal to” type problems by comparing common forms. For example: $\frac{5}{10} = 50\% < 61\%$, so $\frac{5}{10} < 61\%$.

Summary

Percentages are just fractions with denominator 100. They are widely used to compare quantities, find proportions, describe changes, express ratios, and calculate profits, losses, and interests.

- Percentages are written with the % symbol; $x\% = \frac{x}{100}$.
- To convert a fraction/decimal to a percentage, multiply it by 100.
- To convert a percentage to a fraction, write it over 100 and reduce if needed.
- Percentages provide a simple way to compare, estimate and calculate values in many real-life situations.
- Interest rates, profits/losses, growths, and reductions are always best calculated using percentages.

We have also learned to express ratios, increases and decreases, and compound growth in percentages. For compound interest, the total amount becomes $\text{principal} \times (1$

$+ \text{rate})^{\text{terms}}$. This helps solve financial problems more easily. Drawing rough diagrams and estimating answers quickly can help solve questions smartly.

